

Chapter 3

Inner Products and Hilbert Spaces

In a normed vector space, each vector has an assigned *length*, and from this we obtain the *distance* from x to y as the length of the vector $x - y$. For vectors in $\mathbb{R}^d$ and $\mathbb{C}^d$ we also know how to measure the *angle* between vectors. In particular, two vectors x, y in $\mathbb{R}^d$ or $\mathbb{C}^d$ are perpendicular, or *orthogonal*, if their dot product is zero. An inner product is a generalization of the dot product, and it provides us with a generalization of the notion of orthogonality to vector spaces other than $\mathbb{R}^d$ or $\mathbb{C}^d$.

3.1 The Definition of an Inner Product

Here are the defining properties of an inner product.

✓ **Definition 3.1.1 (Semi-Inner Product, Inner Product).** Let H be a vector space over either the real field $\mathbb{R}$ or the complex field $\mathbb{C}$. A *semi-inner product* on H is a scalar-valued function $\langle \cdot, \cdot \rangle$ on $H \times H$ such that for all vectors $x, y, z \in H$ and all scalars a, b we have:

- (a) Nonnegativity: $\langle x, x \rangle \geq 0$,
- (b) Conjugate Symmetry: $\langle x, y \rangle = \overline{\langle y, x \rangle}$, and
- (c) Linearity in the First Variable: $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$.

If a semi-inner product $\langle \cdot, \cdot \rangle$ also satisfies:

- (d) Uniqueness: $\langle x, x \rangle = 0$ if and only if $x = 0$,

then it is called an *inner product* on H . In this case, H is called an *inner product space* or a *pre-Hilbert space*. $\diamond$

The usual dot product

$$u \cdot v = u_1 \overline{v_1} + \cdots + u_n \overline{v_n} \quad (3.1)$$

* Every IPS is a NLS with the norm $\|x\| := \sqrt{\langle x, x \rangle}$ and a metric space with the metric $d(x, y) := \|x - y\| = \sqrt{\langle x - y, x - y \rangle}$.

is an inner product on $\mathbb{R}^d$ (using real scalars) and on $\mathbb{C}^n$ (using complex scalars). Of course, if the scalar field is $\mathbb{R}$, then the complex conjugate in equation (3.1) is superfluous. Similarly, if H is a real vector space then the complex conjugate in the definition of conjugate symmetry is irrelevant since $\langle x, y \rangle$ will be real.

If $\langle \cdot, \cdot \rangle$ is a semi-inner product on a vector space H , then we define

$$\|x\| = \langle x, x \rangle^{1/2}, \quad x \in H.$$

We will prove in Lemma 3.2.3 that $\|\cdot\|$ is a seminorm on H , and therefore we refer to $\|\cdot\|$ as the *seminorm induced by $\langle \cdot, \cdot \rangle$* . Likewise, if $\langle \cdot, \cdot \rangle$ is an inner product, then we will prove that $\|\cdot\|$ is a norm on H , and in this case we refer to $\|\cdot\|$ as the *norm induced by $\langle \cdot, \cdot \rangle$* . It may be possible to place other norms on H , but unless we explicitly state otherwise, we assume that all norm-related statements on an inner product space are taken with respect to the induced norm.

✓ 3.2 Properties of an Inner Product

Here are some useful properties of inner products (the proof is assigned as Problem 3.9.4).

Lemma 3.2.1. *If $\langle \cdot, \cdot \rangle$ is a semi-inner product on a vector space H , then the following statements hold for all vectors $x, y, z \in H$ and all scalars a, b .*

- (a) Antilinearity in the Second Variable: $\langle x, ay + bz \rangle = \bar{a} \langle x, y \rangle + \bar{b} \langle x, z \rangle$.
- (b) Polar Identity: $\|x + y\|^2 = \|x\|^2 + 2 \operatorname{Re}(\langle x, y \rangle) + \|y\|^2$.
- (c) Pythagorean Theorem: *If $\langle x, y \rangle = 0$, then $\|x \pm y\|^2 = \|x\|^2 + \|y\|^2$.*
- (d) Parallelogram Law: $\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$. $\diamond$

The inequality that we prove next is variously known as the *Schwarz*, *Cauchy-Schwarz*, *Cauchy-Bunyakovski-Schwarz*, or *CBS Inequality*.

✓ **Theorem 3.2.2 (Cauchy-Bunyakovski-Schwarz Inequality).** *If $\langle \cdot, \cdot \rangle$ is a semi-inner product on a vector space H , then*

$$\forall x, y \in H, \quad |\langle x, y \rangle| \leq \|x\| \|y\|.$$

Proof. If $x = 0$ or $y = 0$ then there is nothing to prove, so suppose that x and y are both nonzero. Let $\alpha \in \mathbb{C}$ be a scalar such that $|\alpha| = 1$ and

$$\langle x, y \rangle = \alpha |\langle x, y \rangle|.$$

Then for each $t \in \mathbb{R}$, the Polar Identity implies that

$$\begin{aligned}
0 \leq \|x - \alpha ty\|^2 &= \|x\|^2 - 2 \operatorname{Re}(\langle x, \alpha ty \rangle) + t^2 \|y\|^2 \\
&= \|x\|^2 - 2t \operatorname{Re}(\bar{\alpha} \langle x, y \rangle) + t^2 \|y\|^2 \\
&= \|x\|^2 - 2t |\langle x, y \rangle| + t^2 \|y\|^2.
\end{aligned}$$

This is a real-valued quadratic polynomial in the variable t . Since this polynomial is nonnegative, it can have at most one real root. This requires that the discriminant of the polynomial be at most zero, so we must have

$$(-2|\langle x, y \rangle|)^2 - 4\|x\|^2\|y\|^2 \leq 0.$$

The desired result follows by rearranging this inequality. $\square$

By combining the Polar Identity with the Cauchy–Bunyakovski–Schwarz Inequality, we can now prove that the induced seminorm $\|\cdot\|$ satisfies the Triangle Inequality.

Lemma 3.2.3. *Let H be a vector space. If $\langle \cdot, \cdot \rangle$ is a semi-inner product on H , then $\|\cdot\|$ is a seminorm on H , and if $\langle \cdot, \cdot \rangle$ is an inner product on H , then $\|\cdot\|$ is a norm on H .*

Proof. The only property of a seminorm that is not obvious is the Triangle Inequality. To prove this, we use the Polar Identity and the Cauchy–Bunyakovski–Schwarz Inequality to compute that

$$\begin{aligned}
\|x + y\|^2 &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \\
&\leq \|x\|^2 + 2|\langle x, y \rangle| + \|y\|^2 \\
&\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 \\
&= (\|x\| + \|y\|)^2.
\end{aligned}$$

The Triangle Inequality follows by taking square roots. $\square$

3.3 Hilbert Spaces

The question of whether every Cauchy sequence in a given inner product space must converge is very important, just as it is in a metric or normed space. We give the following name to those inner product spaces that have this property.

Definition 3.3.1 (Hilbert Space). An inner product space H is called a *Hilbert space* if it is complete with respect to the induced norm. $\diamond$

Thus, an inner product space is a Hilbert space if and only if every Cauchy sequence in H converges to an element of H . Equivalently, a Hilbert space is

an inner product space that is a Banach space with respect to the induced norm.

Using real scalars, $\mathbb{R}^d$ is a Hilbert space with respect to the usual dot product. Likewise, $\mathbb{C}^d$ is a Hilbert space with respect to the dot product if we use complex scalars.

Here is an example of an infinite-dimensional Hilbert space (compare this to Example 1.1.2, which introduced the space ℓ^1).

✖ *Example 3.3.2.* Given a sequence of real or complex scalars

$$x = (x_k)_{k \in \mathbb{N}} = (x_1, x_2, \dots),$$

we define the ℓ^2 -norm of x to be

$$\|x\|_2 = \|(x_k)_{k \in \mathbb{N}}\|_2 = \left(\sum_{k=1}^{\infty} |x_k|^2 \right)^{1/2}. \quad (3.2)$$

Note that we have not yet proved that $\|\cdot\|_2$ is a norm in any sense; we will address this issue below.

We say that a sequence $x = (x_k)_{k \in \mathbb{N}}$ is *square summable* if $\|x\|_2 < \infty$, and we let ℓ^2 denote the space of all square summable sequences, i.e.,

$$\ell^2 = \left\{ x = (x_k)_{k \in \mathbb{N}} : \|x\|_2^2 = \sum_{k=1}^{\infty} |x_k|^2 < \infty \right\}.$$

We will define an inner product on ℓ^2 . (First we recall that the arithmetic-geometric mean inequality implies that if $a, b \geq 0$, then

$$ab \leq \frac{a^2}{2} + \frac{b^2}{2}.)$$

Consequently, if we choose any two vectors $x = (x_k)_{k \in \mathbb{N}}$ and $y = (y_k)_{k \in \mathbb{N}}$ in ℓ^2 , then

$$\sum_{k=1}^{\infty} |x_k y_k| \leq \sum_{k=1}^{\infty} \left(\frac{|x_k|^2}{2} + \frac{|y_k|^2}{2} \right) = \frac{\|x\|_2^2}{2} + \frac{\|y\|_2^2}{2} < \infty. \quad (3.3)$$

Therefore we can define

$$\langle x, y \rangle = \sum_{k=1}^{\infty} x_k \overline{y_k}, \quad x, y \in \ell^2, \quad (3.4)$$

because equation (3.3) tells us that this series of scalars converges absolutely. In particular, for $x = y$ we have

$$\langle x, x \rangle = \sum_{k=1}^{\infty} |x_k|^2 = \|x\|_2^2. \quad (3.5)$$

We can check that the function $\langle \cdot, \cdot \rangle$ satisfies all of the requirements of an inner product. For example, simply by definition we have $0 \leq \langle x, x \rangle < \infty$ for every $x \in \ell^2$. Further, if $\langle x, x \rangle = 0$ then it follows from equation (3.5) that $x_k = 0$ for every k , so $x = 0$. This establishes the nonnegativity and uniqueness requirements of an inner product, and the conjugate symmetry and linearity in the first variable requirements are easily checked as well.

Thus $\langle \cdot, \cdot \rangle$ is an inner product on ℓ^2 , and therefore ℓ^2 is an inner product space with respect to this inner product. Further, equation (3.5) shows that the norm induced from this inner product is precisely the ℓ^2 -norm $\|\cdot\|_2$, so Lemma 3.2.3 implies that $\|\cdot\|_2$ really is a norm on ℓ^2 .

Is ℓ^2 a Hilbert space? This is slightly more difficult to check, but an argument very similar to the one used in Theorem 1.2.6 shows that every Cauchy sequence in ℓ^2 converges to an element of ℓ^2 (we assign the details as Problem 3.9.9). Consequently ℓ^2 is complete, so it is a Hilbert space. $\diamond$

The space ℓ^1 is a proper subspace of ℓ^2 . For, if $x = (x_k)_{k \in \mathbb{N}}$ belongs to ℓ^1 then we must have $|x_k| < 1$ for all large enough k , say $k > N$, and therefore

$$\sum_{k=N+1}^{\infty} |x_k|^2 \leq \sum_{k=N+1}^{\infty} |x_k| \leq \|x\|_1 < \infty.$$

Hence $x \in \ell^2$. On the other hand, the sequence

$$x = \left(\frac{1}{k}\right)_{k \in \mathbb{N}} = \left(1, \frac{1}{2}, \frac{1}{3}, \dots\right)$$

belongs to ℓ^2 , but it does not belong to ℓ^1 . Now, since ℓ^1 is a subspace of ℓ^2 , we can define an inner product on ℓ^1 simply by restricting equation (3.4) to vectors in ℓ^1 . However, the norm induced from this inner product is $\|\cdot\|_2$ rather than $\|\cdot\|_1$. Although ℓ^1 is complete with respect to the norm $\|\cdot\|_1$, it is not complete with respect to the norm $\|\cdot\|_2$ (see Problem 3.9.10). Therefore ℓ^1 is an inner product space with respect to the inner product defined in equation (3.4), *but it is not a Hilbert space*.

Here is a different example of an inner product space.

*** Example 3.3.3.** Let $C[a, b]$ be the space of all continuous functions of the form $f: [a, b] \rightarrow \mathbb{C}$. All functions in $C[a, b]$ are bounded and Riemann integrable. If we choose $f, g \in C[a, b]$, then the product $f(x)\overline{g(x)}$ is continuous and Riemann integrable, so we can define

$$\langle f, g \rangle = \int_a^b f(x)\overline{g(x)} dx, \quad f, g \in C[a, b]. \quad (3.6)$$

We can easily see that $\langle \cdot, \cdot \rangle$ satisfies the nonnegativity, conjugate symmetry, and linearity in the first variable requirements stated in Definition 3.1.1, and hence is at least a semi-inner product on $C[a, b]$. According to Problem 3.9.8, the uniqueness requirement also holds, and therefore $\langle \cdot, \cdot \rangle$ is an inner product on $C[a, b]$. The norm induced from this inner product is

$$\|f\|_2 = \langle f, f \rangle^{1/2} = \left(\int_a^b |f(x)|^2 dx \right)^{1/2}, \quad f \in C[a, b].$$

We call $\|f\|_2$ the L^2 -norm of the function f . $\diamond$

Although we have defined an inner product on $C[a, b]$, we will prove that $C[a, b]$ is not complete with respect to the norm $\|\cdot\|_2$ that is induced from that inner product (in contrast, note that Theorem 2.9.3 implies that $C[a, b]$ is complete with respect to a different norm—the uniform norm $\|\cdot\|_u$).

Lemma 3.3.4. $C[a, b]$ is not a Hilbert space with respect to $\langle \cdot, \cdot \rangle$.

Proof. For simplicity of presentation we will take $a = -1$ and $b = 1$. As illustrated in Figure 3.1 for the case $n = 5$, define

$$f_n(x) = \begin{cases} -1, & -1 \leq x \leq -\frac{1}{n}, \\ \text{linear}, & -\frac{1}{n} < x < \frac{1}{n}, \\ 1, & \frac{1}{n} \leq x \leq 1; \end{cases}$$

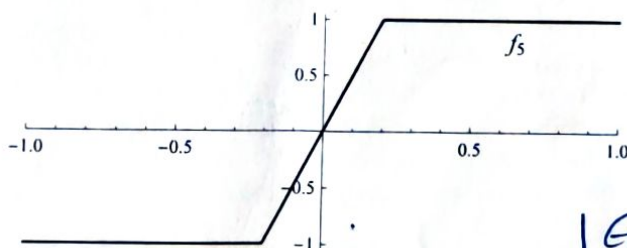


Fig. 3.1 Graph of the function f_5 .

Given positive integers $m < n$ we compute that

$$\begin{aligned} \|f_m - f_n\|_2^2 &= \int_{-1}^1 |f_m(x) - f_n(x)|^2 dx \\ &= \int_{-1/m}^{1/m} |g(x) - f_n(x)|^2 dx \\ &\leq \int_{-1/m}^{1/m} 1 dx = \frac{2}{m}. \end{aligned}$$

Given +ve integer $m < n$,
 f_m and f_n coincide
 outside of $(-\frac{1}{m}, \frac{1}{m})$
 and for $|t| < \frac{1}{m}$,
 $|f_m(t) - f_n(t)|$
 $= |m - n| |t| < |1 - \frac{n}{m}| \leq 1$.
 $\therefore \|f_m - f_n\|_2^2 \leq \int_{-1/m}^{1/m} 1 dt = \frac{2}{m}$
 $\Rightarrow \{f_n\}_{n \in \mathbb{N}}$ is Cauchy.
 The limit f^n is
 $f(t) = \begin{cases} -1, & t \leq 0 \\ 1, & t > 0 \end{cases}$
 which is not continuous.
 Thus $C[-1, 1]$ is not complete.

Therefore, if we fix $\varepsilon > 0$ then for all m, n large enough we will have $\|f_m - f_n\|_2 < \varepsilon$. This shows that $\{f_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence in $C[-1, 1]$. However, we will prove that there is no function $g \in C[-1, 1]$ such that $\|g - f_n\|_2 \rightarrow 0$ as $n \rightarrow \infty$.

Suppose that there was some $g \in C[-1, 1]$ such that $\|g - f_n\|_2 \rightarrow 0$. Fix $0 < c < 1$, and suppose that h is not identically 1 on $[c, 1]$. Then

$$C = \int_c^1 |g(x) - 1| dx > 0.$$

On the other hand, f_n is identically 1 on $[c, 1]$ for all $n > 1/c$, so for all large enough n we have

$$\begin{aligned} \|g - f_n\|_2^2 &= \int_{-1}^1 |g(x) - f_n(x)|^2 dx \\ &\geq \int_c^1 |g(x) - f_n(x)|^2 dx \\ &= \int_c^1 |g(x) - 1| dx \geq C. \end{aligned}$$

Since $C > 0$, it follows that $\|g - f_n\|_2 \not\rightarrow 0$, which is a contradiction. Therefore g must be identically 1 on $[c, 1]$. This is true for every $c > 0$, so $g(x) = 1$ for all $0 < x \leq 1$. A similar argument shows that $g(x) = -1$ for all $-1 \leq x < 0$. There is no continuous function that takes these values, so we have obtained a contradiction.

Thus, although $\{f_n\}_{n \in \mathbb{N}}$ is Cauchy in $C[-1, 1]$, there is no function in $C[-1, 1]$ that this sequence can converge to with respect to the induced norm $\|\cdot\|_2$. Therefore $C[-1, 1]$ is not complete with respect to this norm. $\square$

Each incomplete normed space that we encountered prior to $C[a, b]$ was contained in some larger complete space. For example, c_{00} is incomplete with respect to the norm $\|\cdot\|_1$, but it is a subset of ℓ^1 , which is complete with respect to $\|\cdot\|_1$. Likewise, $C_c(\mathbb{R})$ is incomplete with respect to the uniform norm, yet it is contained in $C_0(\mathbb{R})$, which is a Banach space with respect to that norm. It is likewise true that the inner product space $C[a, b]$ is contained in a larger Hilbert space, the Lebesgue space $L^2[a, b]$. However, defining $L^2[a, b]$ is beyond the scope of this manuscript, as it requires the theory of the Lebesgue integral. We refer to texts such as [Fol99], [SS05], or [WZ77] for discussion of the Lebesgue integral and the Hilbert space $L^2[a, b]$. ✓